Lath

Squads v4 answer verification

Verify the Lath agent loop that answered it: every step recorded on Solana, every model call paid on Tempo.

Lath · agent squads-v4-review · loop loop-b37ba7c46db86bd4 · Solana devnet + Tempo mainnet · 2026-10-12

Question. Before I approve a Squads transaction: who can upgrade this program, and what must pass before funds move or members change?

Answer · goal reached at step 4, Jev judged it achieved (0.98)

Before you approve a Squads transaction: a Squads program upgrade can only be done by the Squads-team keypair that holds the loader upgrade authority with no multisig vote, while moving funds, changing members, and setting spending limits all run through the multisig's own config/proposal flow, and spending limits let listed signers move capped amounts without a vote.

`4 steps · 20 paid model calls · 0.0599 USDC charged on Solana · 0.0547 USDC.e paid on Tempo · receipt on SAS`

Everything below opens in a browser, with nothing to install: Solana Explorer (devnet), explore.tempo.xyz (Tempo mainnet) and the Irys gateway.

THE RUN, IN ORDER

1	**Sources pinned before the run** GitHub @ af94153 · Irys index · Squads program	The Squads v4 instructions folder at commit af94153, and the program on Solana devnet at slot 510428403. The Irys index is a JSON list of all 26 files the agent could read, each with its sha256; the sandbox had no network, so these are the only sources.
2	**Agent registered on Solana** registration tx	A successful transaction to the lath program AzAe...BKKz at slot 510426757: the agent's own hash chain of records starts here.
3	**Loop key authorized on Tempo** authorize tx · the key	Sent by Lath's treasury 0x18fB...e0B0 to Tempo's AccountKeychain: key 0x8Bae...1aDc may spend at most 1.15 USDC.e (the 1 USDC budget plus Lath's 15 % decision share), until 2026-11-11. Tempo enforces that limit: a payment past it is refused by Tempo, not by Lath's code. The key's page lists its authorize, the channel it signed, and its revoke.
4	**Payment channel opened** open tx	An MPP payment channel on Tempo's escrow from the treasury to OpenRouter's gateway, signed by the loop's key. Each model call after it is a signed voucher on this channel, not a transaction.
5	**Four steps, one part of the answer each** step 1 · step 2 · step 3 · step 4	Each link is the step's loop.step checkpoint record on the agent's chain. The table below gives each step's calls, costs and the decision model's verdict, which the record commits.
6	**Answer stored on Tempo** storage write	A 0.000200 USDC.e payment to Tempo's storage gateway for report/note.md (4,197 bytes). Its sha256 is committed by a storage.put record on Solana at seq 120.
7	**Channel settled** close tx	The channel closes with the last voucher: 0.053870 USDC.e to OpenRouter's payee 0xca4e...79Fe, 0.113959 USDC.e refunded to the treasury.
8	**Key revoked** revoke tx	The treasury revokes the loop's key at the stop: from here it can sign nothing, whatever was left of its limit.
9	**Receipt on SAS** attest tx · attestation	An attestation on the Solana Attestation Service. Its data: agent 877662...a212, loop loop-b37ba7c46db86bd4, last record seq 114, chain head 9b308d...b3b3, goal hash 8c0c0e...ffdd, 4 steps, 0.059910 USDC charged, 0.053870 USDC.e spent on Tempo, Tempo payer 0x8bae...1adc.

THE FOUR STEPS

STEP	PART WRITTEN	CHECKPOINT	CALLS	CHARGED, USDC	PAID ON TEMPO, USDC.E	DECISION	NOTE STORED
1	upgrades	seq 40	4	0.013158	0.012002	continue (0.98)	1,160 B
2	moving funds	seq 65	5	0.014783	0.013483	continue (1)	1,916 B
3	changing members	seq 93	6	0.019398	0.017691	continue (1)	3,280 B
4	spending limits	seq 114	5	0.011691	0.010694	achieved (0.98)	4,197 B

HOW THE PIECES TIE TOGETHER

- The key authorized in row 3 (0x8Bae...1aDc) signed the channel in row 4, is revoked in row 8, and is the Tempo payer the receipt names.
- The four steps' Tempo payments add up to 0.053870 USDC.e, what the close in row 7 paid OpenRouter and the receipt's Tempo total.
- The receipt's USDC total (0.059910) is the four steps' charges (0.059030) plus the four note writes (4 × 0.000220), each a debit from the owner's capped allowance on Solana.
- The receipt's seq 114 is step 4's checkpoint record, the record the decision model's achieved is committed in.
- The answer's first claim can be checked directly: the Squads program's page shows its upgrade authority, HM5y...dtvK, a single key, and the source at af94153 hard-codes that key as INITIALIZER (program_config_init.rs, line 10).

Squads program on Solana Explorer · INITIALIZER at af94153 · Lath's treasury on Tempo

What this proves: what the agent read, what it did, what each step cost and who paid, and that none of it changed since. It does not prove the answer is right; every claim in the note cites a pinned file you can open.